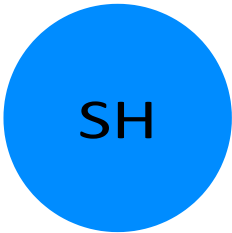


SHANTANU HULE

Python Automation || MLOps

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SUMMARY

I am an AI/ MLOPS Engineer with a strong foundation in cloud technologies and machine learning. My expertise includes managing and optimizing CI/CD processes, data validation, and deploying complex web applications using modern frameworks. I've contributed to various Aldriven projects, from health diagnostics to cloud-based anomaly detection, demonstrating my ability to elevate operational efficiency. I am passionate about continuous learning and leveraging technology to solve real-world problems

EXPERIENCE

SME MLOps

Jio Platforms

03/2025 Present

Company Description

- Designed and implemented **PowerShell, Bash, and Python automation scripts** for Kubernetes tasks such as deployment backups, ZooKeeper config snapshots, and DR environment config management.
- Integrated Prometheus metrics with Grafana to visualize API performance, traffic analysis, and error trends.
- Developed **alerting scripts** using PowerShell and Python for Kubernetes-based applications to detect failures
- Automated **PVC migration processes** across environments, ensuring zero-downtime transitions and consistent storage configurations.
- Performed **UAT, production, and DR deployments**, including config updates, CRP validations, and rollout monitoring via Azure Pipelines.
- Created a **monitoring microservice** to execute database queries and analyze logs for early anomaly detection in application behavior.
- Integrated **Azure DevOps Pipelines** for config deployments, code changes, and environment-specific automation workflows.
- Analyzed failures in the **Azure Portal Failure section**, traced errors to Kubernetes logs, debugged issues, and coordinated with development teams for resolution.
- Nginx Load Counter Prediction & Auto-Scaling Trigger**: set up a prediction-based scaling mechanism instead of CPU-based scaling, leading to 20% more efficient resource usage.
- Log-Based Anomaly Detection for API Errors**: integrated anomaly detection into our logging pipeline, enabling proactive incident alerts before customer impact
- Container Resource Usage Forecasting: Instead of reacting to resource saturation, we implemented predictive scaling based on ML forecasts

KEY ACHIEVEMENTS

- OpenShift Deployment**
Successfully deployed a web application on an OpenShift cluster, thoroughly optimized for performance and security

- AI Application Development**
Created advanced web applications for various AI projects, showcasing expertise in innovative tech solutions

SKILLS

Python	AWS	Azure	Django
Flask	Selenium	SQL	CICD
Data Science	ML	AI	
MLOPS	MLflow	FastAPI	
Docker	Kubernetes	Helm	
Jenkins	Prometheus	Evidently	
Grafana			
Devops	Git	Openshift	
Deployment	Change management		
CICD	Sanity Testing	Linux	
Bash			

EXPERIENCE



AI/ MLOPS Engineer

Jio Platforms

📅 12/2023 03/2025

Company Description

- Developed and deployed AI-driven automation solutions for monitoring and optimizing IT operations.
- Designed and implemented Grafana dashboards for real-time monitoring of Kubernetes workloads, application traffic, and system health.
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- Built and maintained automation frameworks using Python, Django, FastAPI, and Selenium.
- Designed and implemented Kubernetes-based deployments for scalable applications, managing ingress configurations, services, and load balancing.
- Created API flow mapping tools using Django (backend) and React (frontend) for better visualization of application architecture.
- Developed and optimized web automation frameworks using Selenium and Appium for automated testing of web and mobile applications
- Integrated Azure DevOps APIs (v5.1 for CI/CD pipeline automation and workflow optimization.
- Deployed machine learning models and data science pipelines for predictive analysis in system performance monitoring
- Troubleshoot and optimized Nginx, OpenShift routes, and Kubernetes networking for secure and efficient API and application access.
- Worked with MongoDB and SQL databases, ensuring efficient data processing and storage solutions.

Machine Learning Intern

Zerot.ai

📅 08/2022 03/2023

A tech startup focused on AI and machine learning solutions

- Conducted time series analysis for oil well performance tracking
- Performed anomaly detection for Google Cloud data, identifying behavioral anomalies across multiple cloud services
- Engaged in feature engineering using GCP virtual machine metrics
- Developed ML models and Azure functions for IoT sensor performance evaluation

Artificial Intelligence Intern

Nothink Technologies

📅 04/2022 08/2022

Company specializing in data-driven marketing solutions

- Utilized data analysis techniques for Facebook campaigns, optimizing targeting and campaign performance
- Conducted time series analysis on various platforms, including Google, LinkedIn, and Facebook

Artificial Intelligence Intern

Rainet Technologies, Delhi

📅 08/2021 03/2022 📍 Delhi

Technological solutions in the AI space

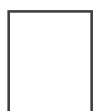
- Performed data analysis for a travel agency, predicting customer claims using machine learning
- Contributed to research paper writing for a medical image classification project
- Applied natural language processing using transformers for feedback classification
- Developed a system to predict customer active and inactive status for a bank

PROJECTS

Healthcare AI Projects

Personal projects focusing on health-related AI applications

- Developed a brain tumor detection model using the darknet framework to predict tumor locations
- Created a chest disease classification model using transfer learning, accessible via GitHub
- Built a Django web application that integrates multiple deep learning models for disease detection



EDUCATION

Bachelor of Engineering

Marathwada Mitra Mandals institute of Technology

07/2019 07/2023

